

Network used for communication.

Communication broken into layers.

Benefits of Layer

1. Each layer have common things (defined function) with a standard.
2. Standard public competition among and layer
particular layer & or every layer a competition.
3. Every layer is change with layer & affect.

Protocols means Rules.

* All communication are governed by protocols.

Standards is Strict universal Rules.

* Made by a standard organization.

* Benefits →

- ① competitive market and open.
- ② compatibility and interoperability.

* Categories →

- ① De facto → By fact → standard organization by fact.
- ② De jure → By law → standard organization, standard law.

Why org. fixed -
st. org. fixed -
standard - fixed.

③ open standards → fixed -
use - fixed.

④ Proprietary standards → fixed -
particular organization & fixed.

ICANN → Assign IP address, Domain names, port numbers

Internet draft → কোনো প্রকট rule test করার জন্য,
IETF এ পর্যালোচনা, এই official status প্রাপ্ত
Rule for Internet draft গুলি

RFC - 791 → Documentation Number.

Protocols are not dependent upon any specific tech.

→ প্রকৃত language এ build করা browser same
protocol follow করে, Like (https) → এর সাথে
- কাজ করে

→ এই নথি
OSI Reference Model (Number matters) → Devised by ISO
TCP/IP conceptual layers (War time এ
- সময়কাল) → Governed by IETF

An people seem to need data processing → code to remember
(7-3) → hardware software, (1-2) → hardware. OSI model
Data Link → hardware + software combination.
An exchange using the OSI model

① Pen testing → Vulnerability testing.

② live server 1 test করতে যা, staging 2 test করতে

④ PicoCTF

④ Live competition

④ BugCrowd solve

③ Presentation Layer →

CSE 421

Translation → 0/1 থেকে A, B, C যাঃ A, B, C থেকে 0/1 বানানো

Compression → size কমানো

Encryption → Data সুরক্ষা / secure করা

④ Session Layer

→ session → Active থাকে, Active না থাকলে logout.

→ It handles

→ initiate dialogs

→ keep active

→ disrupt restart the session for being idle.

* session, presentation, transportation → এটা জিনিস না সব ভেঙে কিনা, layer Application layer → সবকিছু handle করে.

① Transport Layer → যখন থেকে network এর জিনিস শুরু

→ এ, এই layer ৩০ one process থেকে another process

এ message পাঠানো

② Data small piece এ পাঠানো কারণ এটা ছোট

→ এটা না এটা একেবারে, এখানেই করা পাঠানো

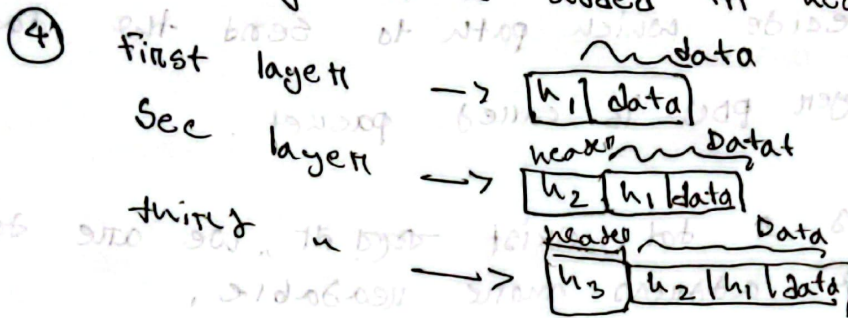
→ এটা, এই small piece এ ভাগ করা

segmentation

Transport layer PDU is called segments

Function →

- ① Segmentation (senden ভাঙবে) and Reassembly (received একত্রে)
- ② Add port (যদি এই ভাগে ভাঙলে কোন নাকি কোন (সবচেয়ে) Sequence Number (এই ভাগে - অন্য এক এক নম্বরে ভাগে আসবে)
- ③ Everything will be added in header section.
- ④



Port number identifies the application

- header 1 16 bit ভাবে থাকে
- Represent it in one decimal

* Transportation layer also maintain Established secure connection.
-সবচেয়ে sender check করে if the receiver is available or not.

* Controls flow of data.

* receiver এর buffer size কম হলে overflow হতে পারে, overflow -যাতে না-হয় (হয়তো) receiver sender কে flow control করে, অন্য (অন্য) যা-

* Error control by acknowledging and resending data.
* -যদি data packet missing -হয় তাহলে receiver -ভাগে

* Multiplexing $\rightarrow$ process where two or more signals are combined into one signal over a shared medium.

* Network Layer :-

It transports data from source to destination.

* Adds IP addresses of sender and receiver.

* Decide which path to send the data.

* Network layer PDU is called packet.

* TCP/IP

* IP addresses do exist. We are doing it to make the address more readable.

* Difference between router and switch.

Switch is used in local area network to use data transfer.

Router is used for WAN.

* Data Link Layer $\rightarrow$ known as data frame.

* 48 bits

* Unique given by manufacturer.

* Link layer frame transfer.

* Link switch considers as a router.

* physical layer $\rightarrow$

* physical topology.

* Transmission mode.

* Specific address → Email address / URL → Falls under in Application layer.

① IP address / Mac address port is stored, not on device.
IP and mac stored in Lan port.
② Mac address hop to hop change.

CSE340 (Chapter - 01)

* etc.

* personal computer → Normal

* Server

→ Huge memory, communicates through network.

* Super

→ Normal computer for special sector & powerful - statistic, like image processing & GPU max.

* Embedded

→ Arduino, pi

questions.

① Understanding Performance → [Algo, Language, compiler, processor and memory, I/O system.]
→ software related.
② high → Assembly → Machine hardware related.

③ processor → ALU, register, etc.

④ Parallelism → multiple process same time & order.

⑤ Pipelining → each processor to multiple component
total time & use of multiple component simultaneously same time & turn.

⑥ Linux & number of instructions - stat.
Application level & work flow describe.

Application Layer

Application Layer
 " " protocol → HTTP
 " " Software → Browser

Client/Server → একটি common device থাকে, এটা থাকে যাতে
 # P2P → এখানে ক্লাস্টার server থাকে না, প্রতিটা মিনি প্রতিটা নিজে
 -সবই connected.
 > এখানে client এবং server কে request করে, then info
 থাকে, এখানে server is always server and client is always
 client.
 > এখানে server/client role switch করে based on need.

URL → protocol : // hostname . domain-name . top-level-domain /path/file

website enter করে এখানে http protocol এর through to
 web server এ hit করে then web server http response
 -করে

http sends two kinds of message

① HTTP request

First Line. ←

- * CR → carriage return character → এর mean
- * LF → line feed character → এর mean
- * Method → to declare what I am doing
GET, POST.
- * If → go to next line.
- * SP → which protocol we are using (gap)
- * Version → HTTP এর status version use
-করে

② 2nd line

→ Header line → Extra information sending to the server.

Host →

③ Entity → More info, Ex → if we submit form to send extra info.

④ HTTP Method type :-

① HEAD → Server will respond without giving the actual content. It will response with 'YES, The page is here'.

② Uploading Form

→ post method → server 2 form data store.

→ URL method → Google Search.

④ RESPONSE MESSAGE

* First line will be status line

① version → http version

② sp → separator

③ status code → 200, 400

④ phrase → 200 OK like
404 Server not found

HTTP connections

- First it will do a TCP request. Which will make sure that server is ok.
- Second server return ok and tcp ready
- Third GET request.
- Fourth server send data
- Fifth connection close - server & server
- Sixth
-

RTT time

→ Small packet request at server -> time totally not matter

* Time to transmit file → website as main file retrieval -> server & server time matter

Non persistent connection

port close -> means file transmit ->

byte → File Size
bit → Speed

Ex from slide

$$RTT = 25 \text{ (requesting Given)} + 25 \text{ (Getting Assuming)} = 50$$

$$\text{Obj. Size} = 10 \text{ MB} = 10 \times 1024 = 10240$$

$$\text{Speed} = 100 \text{ Mbps}$$

$$\frac{100 \text{ Mb}}{10240} = \frac{1 \text{ s}}{10240}$$

PTC

Persistent HTTP $\rightarrow$ Multiple object can be sent over one single connection.

Obj = RTT + (RTT + file size) $\times$ no. of obj

> if all the file size are same.

$$= RTT + (RTT \times \text{no. of object}) + \text{obj. file time}$$

Pipelining did not improve the efficiency. so did not use pipelining.

[to quit protocol]

CSE 340

CPU Time = (no. of CPU clock cycle $\times$ clock cycle time $\times$ number of instruction)

CPU clock cycle $\times$ number of instruction

CPU time = (no. of cycle $\times$ no. of instruction)

num of cycle

Clock Rate = $\frac{\text{clock cycles}}{\text{CPU time}}$

= $\frac{\text{clock period}}{\text{seconds}}$

= $\frac{1}{\text{Clock cycle}}$

CPU time

Stateless → Not keeping state. Server will not keep any data about user. Server will not keep track about user. HTTP is stateless. HTTP stateless because it is complex, needs more memory. But state is not error, that is why cookie came.

Cookie → Four components

1. Cookie header line

3.

4.

* Server cookie id only that communication is

that. Server cookie id save that. It is

client to save browser cookie id save that.

After, After setting cookie id, every

HTTP request will send with the

cookie id after that.

Cookie Used For

1. Auth
2. Shopping cart
3. Recommendation
4. User session maintained by cookie.

With cookie, site is keeping state.

Cookie permit sites to learn a lot, might send email and name to others.

[Web Cache on Proxy Server]

Proxy Server (web cache) →

1. Request goes to proxy server
2. Proxy server if proxy has it, won't hit the main server
3. Client is closer to proxy server

proxy server doesn't have original server

Proxy servers are installed by ISP.

Benefits —>

1. Reduce response time.
2. Saves bandwidth
3. Using proxy server, it can block unwanted traffic or create log about user.
4. Poor 'content provider' effective — ISP ko rokna jaye — website ko user ko server 1 hit, rokate — jaye, known as dense with cache.

Delay — 20 — 30 sec karte karte

$$ART = \underbrace{Lan\ delay}_{\downarrow \text{ latency}} + \underbrace{Access\ delay}_{\downarrow \text{ Lan (server) actual internet (actual internet delay)}} + \underbrace{Internet\ delay}_{\downarrow \text{ actual internet}}$$

* Delay in min is Bad.

* Link Speed — jate jate — jate too costly.

* Average hit means — karte karte jate local server 1 jate karte.

$$* \text{Total response time} = (\text{PRXY hit} \times \text{delay}) + \text{internet hit} (\text{delay} + \text{Access delay})$$

↓ fixed

[Stale Cache]

- * Stale means — jate, jate Updated cache store karte.
- * HTTP request msg —> client req karte wo LAN server 1 req to — jate —> LAN server req jate main server 1 — jate —> Page updated — jate (jate) —> Update — jate —> Client main server 1 update — jate Full website lan 1 jate (200 OK) other wise update — jate (304 OK) jate —> Lan server —> Lan client to jate.

-2ମେ ଫଳ -2ଟି, ଗୁଣିତ ମାନ 20 ବାଲେ 10 -ଗୁଣିତ

DR - 150 - 500 7500 / 1209 11000 2 1100000 9100 8909

45- 51850- 0750 ~~45- 51850- 0750~~ CSE 496

⑦ secret * f. the for range of covered

from article where id = -1 + union + all + select

Ans: $\frac{1}{2}$ band dec $\leftarrow$ 89 AMI, 909, 711 + 1, 2, 3.

① select * from article where id = -1 union all select *

database() --

① group_concat (Table name) [AS] AS alias

#) table - schema - 'Sal' - information - Schema - Table

Sal injection

CSE421

Electronic Mail 1/10/97 9:14:22 (S)

③ Three major component

1. User Agent → the app used to compose and send email

2. mail Setvett $\rightarrow$ it contains mailbox to state mail

message we've for outgoing mail.

if I send mail to multiple server →

SMTP Server

SMTP $\rightarrow$ 1. Send mail from

— 02 09 TCP, from client to server. Using

2. SMTP GET TCP handshake

SMTP Handshake - 4 steps

10/14/2014 - 10/14/2014

- * Recipient uses POP3/IMAP to read the email.
- * Comparison with SMTP and HTTP
 1. SMTP server → mail push / Server to Server possible, POP3 & server mail pull → use server
 2. HTTP server to server not
 3. HTTP pull-push → possible
 4. HTTP, POP, IMAP → web based & mail download

* POP3 vs IMAP

- ① POP3 → through PC to access mail, then local server delete mail, IMAP → server & server & delete
- ② POP3 → can't create, delete or modify mailbox on the mail server. IMAP → can
- ③ SMTP port 25

* POP3 use → one time message to use, so security purpose & use

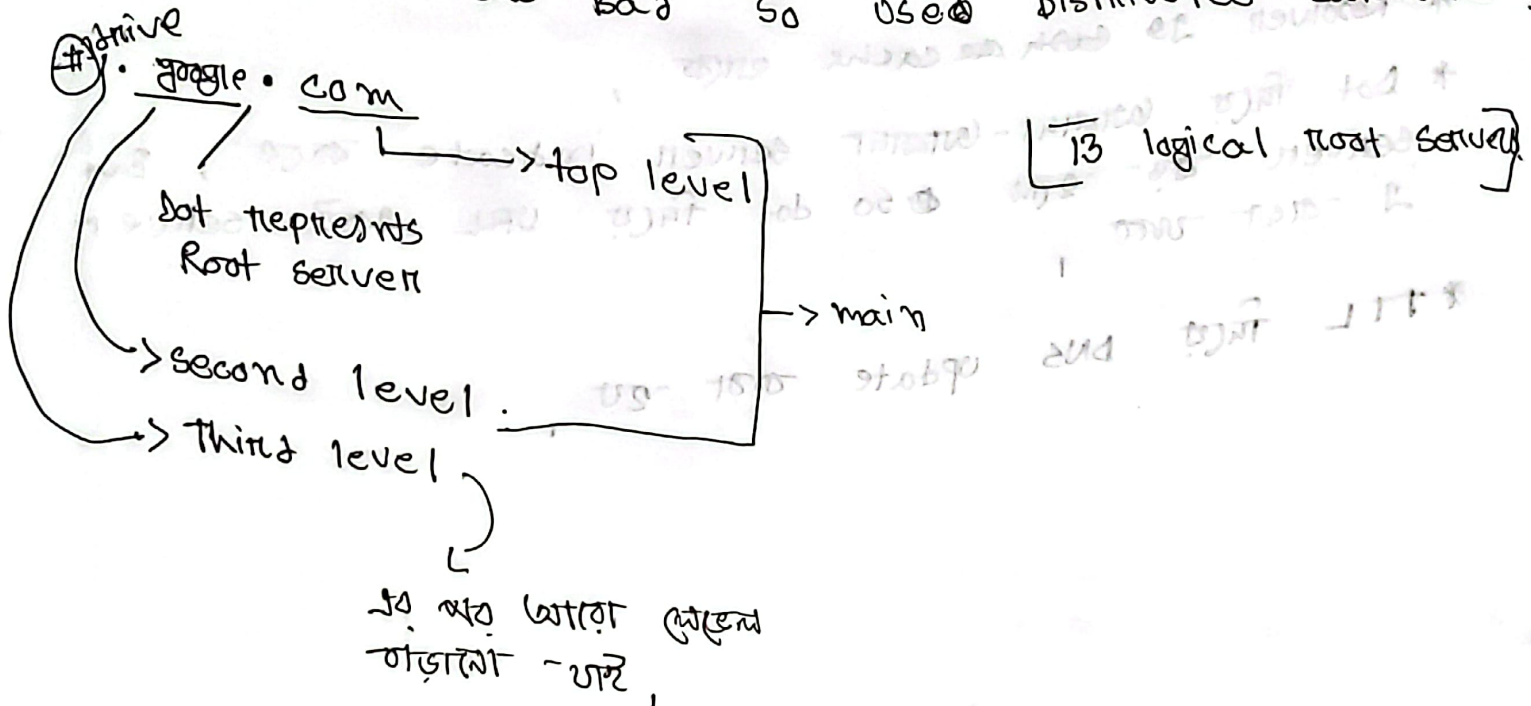
* # DNS

- ⇒ Domain name to IP address
- ⇒ DNS server access → PC to DNS Resolver server to access DNS Resolver

⇒ Does DNS server ~~also~~ do TCP handshake?

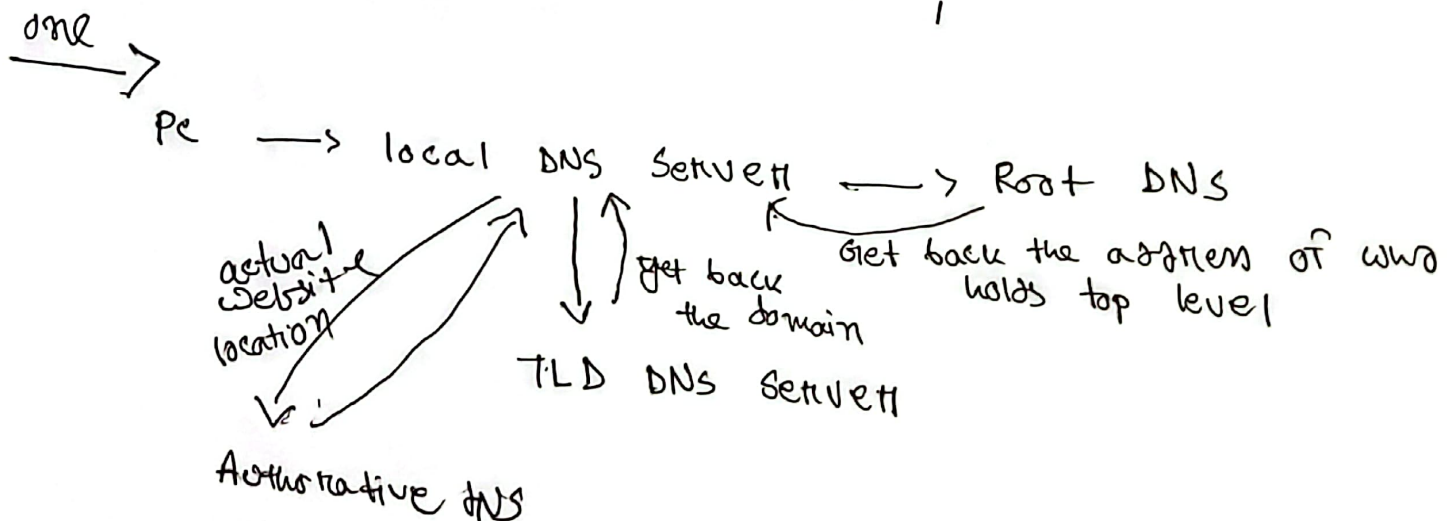
→ Depends on how you set it up.

④ Centralized DNS Bad So used distributed database



(*) Root server —> .com , .org ইত্যাদি টপ লেভেল অর্থাৎ
IP বাধে,
এ প্রকার ইন্টারনেট সার্ভার ০ কোন top level server এ
যাচ্ছে।

① top then top level 1 - 25/70



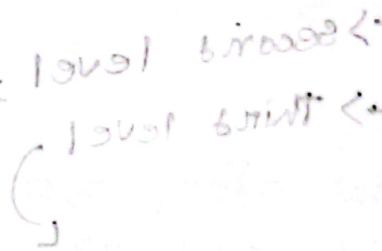
Recursive $\rightarrow$

ਸਰ੍ਹਲ-ਜਿਥੇ headache ਕੋਈ root ਸੈਨਰ ਨਹੀਂ, ਜਿੰਨਾ ਵੱਡਾ
ਜਿਥੇ ਜਿਥੇ ਹੁਣੇ ਪ੍ਰੈਸ਼ਰ ਹੈ

* Resolver JS ~~can~~ cache ~~data~~

* Dot দিয়ে ওয়াশাট-ওয়াশাট servern indicate করে, But এ-দিক-দিক So dot দিয়ে URL গুলো server

* TTL निर्देश DNS update करत-रु



may be total or partial
Syr - total

Heute + 109 (10)

CSE 421 (DNS Queries)

Quiz - 4

Topic - DNS + Email

Wednesday
3rd July

- * DNS Records → DNS server record store,
 format: (name, value, type, ttl)
 # CName → Canonical Name → same website a - [facebook.com](https://www.facebook.com)
 # TypeA → Actually has the ip address and hostname
 # NS → dns server - indirect server, which dns a site is
 # MX → Mail server dns
 * DNS Security → Dos → Denial of Service
DDos → Distributed
 → Multiple pc attack
 → Zombie attack (virus spread - multiple pc capture server & control them)
 Req. (server, client do not know)
DNS cache poisoning → attacker cache -
 which ip address change - server - domain
 name is against a, so people gets redirected
 to the attackers website

— செய்தி —
காலம், இடம், காரணம், விளைவு

$(15x) + 2 \rightarrow 2x + 62$

Question - 01

(a) Here, $f(x) = \tan(x)$

$$P_3(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3$$

for, $\tan(x)$

$$f'(x) = \sec^2 x$$

$$f''(x) = 2\sec^2 x \tan x$$

$$f'''(x) =$$

[CSE421]

Transport layer - কাজ করে একটি process থেকে অন্য process এ ডেটা - বার্তা, port.

process to Network + Network to process

Segmentation using Sequence Number

Multiplexing → একসাথে discord, web browser চালালে

5, 6, 7 to make sure data বিকলিত পৌঁছাইছে না কি

TCP reliability ensure করে - UDP করে না

→ ইচ্ছা করে reliability provide এর জন্য

TCP like email, web browser → এখানে হলে প্যাকা

UDP IPTV, Gaming → কিছু এখানে (সমস্যা প্যাকা নাহে -

HTTP reliability maintain করে না, এটা UDP use করে, Application layer এ reliability maintain করে

UDP reassembly করে না কারণ Sequence নম্বর নাহে

checksum to check if data বিকলিত (হলে কিম্বা

$$5+6=11 \rightarrow 11 \text{ টি পাঠাবে}$$

$$5+7=12 \rightarrow$$

এটা দেখে পাঠাবার error

me don't want Me watching start
cold morning People enjoying rain
ever seen knowing that if I
if there, so will give

Well know $\rightarrow 80, 443 \rightarrow$ fixed port
Registered $\rightarrow$ IANA app or port register
client side a random port (dynamic) $\rightarrow$ use & change

$\rightarrow$ different same port $\rightarrow$ select request that why user process a randomly dynamic port assign

Socket address $\rightarrow$ IP address: port address
 $\rightarrow$ Data transfer socket & network

Multiplexing $\rightarrow$ Data collect and transfer
Demultiplexing $\rightarrow$ Data transfer and process a network
 $\rightarrow$ same socket
UDP demultiplexing identify data through to
(s.IP, s.Port, D.IP, D.Port)

TCP demultiplexing identify data through to
(s.IP, s.Port, D.IP, D.Port)
 $\rightarrow$ creates different socket

#

⑩ Misc

- Password-leaf db (Dehashed)
- Git (config file, informed attack)
- CMS (WordPress - wpscan (List of users + plugin), Joomla, Drupal)
- Subdomain takeover
- Automated Scanner (Manual checking important)

CSE 421

Byte Number: →
for computer generated random numbers that could be in between (0 and $2^{32}-1$)

first byte
1067



last byte
4066

1067 included that's why 4066 No 4
4064

Sequence Number → ~~sequence~~ first byte of every sequence number

Acknowledgment → received next byte of data expect

Suppose 1 - 10 are sent, receiver will send 11 as their acknowledgment number.

* Also known as Exceptional acknowledgment number available which will provide the first available - after that one.

- # TCP setup a connection using three-way handshake.
- Sender →
- # to start a connection, the sequence number will be anything made by OS.
 - # Acknowledge bit will be 0, as we have sent nothing while making this connection.
 - # S will be on (sync bit)
- receiver.
- # Sequence → 0 (sequence number device based, for this device sender is 0)
 - # Acknowledge → 1001 (client is 1000 and receiver is 1001)
 - # A, S will be turned on
- Sender →

- # Sequence → 1001 → data
 - # Acknowledgment → 1 → (1 is data, expect round → server/sender capacity and establish is, wait - wait, connection)
- # Half close :

Reliability in TCP

- ① Error control → slide
- ② Acknowledgement → Acknowledgement do not consume sequence number.
- ③ Retransmission → timer maintain - wait, 500ms wait - wait,

Retransmission $\rightarrow$ client packet ^{server} acknowledgement
not received, wait - timeout, if ack not received (timeout) -
packet is retransmitted.

Multiple ack same request is not allowed.

Missing packet then receiver ack request - error - discard - discard.

duplicate package received then - ack not received.

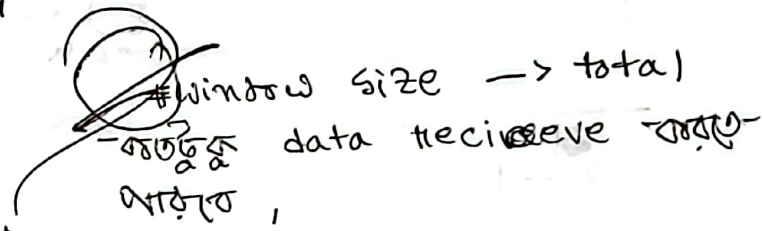
Backs fast Retransmission $\rightarrow$ receiver starts timer (T) -
until Retransmission time exceed T then -
transmit, then timer desired packet received.

Packet drop: - receiver for - error.

Sender sliding window:

$S_n \rightarrow$ first outstanding byte

$S_n \rightarrow$ Next byte to send.



What is window size and sliding window?

100 byte consumed $\rightarrow$ 100 byte Application layer
data received. So window size will increase. as the
the ~~new size~~ size will be - total,
100 byte

sliding window protocol.

① * Go back N ARQ $\rightarrow$ receiver (window) data -
received. then - receiver - data accept -
transmit. - receiver - data accept -
transmit.

* NAK $\rightarrow$ Negative acknowledgement, -
not data -
data -
data -

② * Selective Repeat ARQ $\rightarrow$ -
data -
data -
data -

[CSE421]

Network Layer : IP Addressing

- # কোন লগ দিবে - যাবে, কোথায় আসবে, কোথায় থেকে যাবে, কত
- # $192.168.10.1$ → যেই network এ থাকলে, ওই Network এর unique device সূত্র identify করা,
 Network part Host part
 → ইতি format fixed না, change হতে পারে,

Learn Binary to Decimal

- # we give ip address to the port. - যতগুলো port, ততগুলো ip address.
- # Router creates separate network
- # switch এ ip address নাহলে
- # ip address কোন use করে, MAC address কোন ইডিফাই করে না?
- # 255 same address Network এ connect করা - পারে, 8 bits. → যদি এর বেশি - কানেক্ট করা নাহলে তাহলে subnet Mask এ amount of host change করতে হবে,
- # Subnet Mask / prefix Mask
- # Prefix Mask tells us How many octets are the part of Network. ~~part~~
- # $255.255.255.0/24$ → কিন্তি octate এ 255 - হতে,
- # Subnet Mask (প্রতি- prefix Mask on vice versa conversion)
- # $10.24.36.2/12$ →

255	240	0	0
11111111	11110000	0000	0000
8	4+4	8	8
Network			
Host			

- # Network address → Here all host bits are zero
- # Broadcast → Here all the host bits are zero
- that means ইতি (সংকেত) - সবার - সংকেত - যাবে,

1957

19. 10. 1941

01 821 201

1944

1007 7-10-55 - 1007 24 of 1007 24

⑤ Mini militia $\longleftrightarrow$ App

$\frac{1}{2} \rightarrow$

present with

11/10/2014

Handwritten: 100M Hard

① कबूतर flexa २ अक्षर १० ७७

ପ୍ରଶ୍ନ Hexa A ମାତ୍ର ମୂଲ୍ୟ

etia teon no

526128 e. i. ' ਤਿਸਾਨਾ ਮਾਨ ਮਾਨ

(A57) 0 : 0 : 0 : 0 : A ————— স্বপ্নের সিন্ধি গল্প

Function of Network Layer

1. Forwarding :- Value of Router (যাও ১ টি Router) -> Right Router
2. Routing :- Choose Path (যাও ১ টি Path) best path

Network Layer

Reliability provide. So we do not need to make the connection. Faster.

Virtual Circuits

Like Mobile phone -> One to One path create. When i am calling someone.

Network Layer

ICMP is a message transfer. ICMP protocol invoke. ICMP also used on ping.

Header len -> Min value : 5, Max value : 15. as it has only 4 bits.

$$5 \times 4 = 20 \text{ (min)}$$

$$15 \times 4 = 60 \text{ (max)}$$

Type of Service

-> for service. -> data to wireless network.

Time to live -> hop count. Time to live -> hop count. Time to live -> hop count.

- # IP Fragmentation - $\rightarrow$ data size ko chhota karke bhejenge,
 Don't flag bit $\rightarrow 1$ router ko data chhota karke bhejne hai,
 - agar router ko data chhota karke bhejne hai to router will drop it and send a message to the user about this drop using ICMP.

Data size ko chhota karke header size include.
 if MTU 1500 then we will be able to send max 1480 data cause min 20 header will be there.

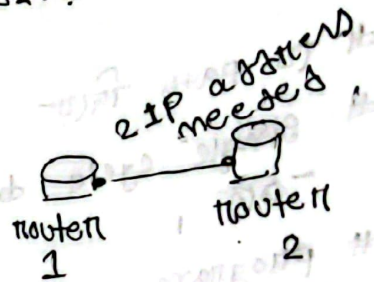
Why are we dividing sequence number by 8? $\rightarrow$ Because of sequence number, 32 bits but offset number 13 bits, that's why.

IPv6 HL fixed $\rightarrow 40$ bits $\rightarrow$ that's why fixed hai,
 # IPv6 fragmentation hai $\rightarrow$ ICMP6 se message milega.

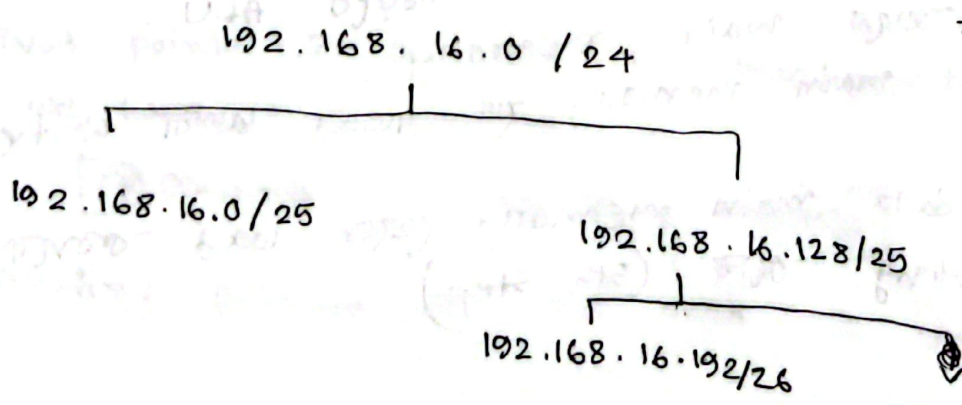
Protocol and options $\rightarrow$ Next header
 $\rightarrow$ data part $\rightarrow$ data part.

GSE421

- # Subnetting → Enough ip নাহি দেওয়া - মূল, তাই subnet বানাচ্ছি।
- # Network ip → শত দেওয়া - যারনা subnet করা - সময়, we will use leftover host address to create this subnet.
- # Router এর প্রতিটি interface এককটা network ~~করা~~ , Represent করে।
- # 120 - হল - host address - হল subnet করা - পরিমাণ না।
- # 192.168.10.0/27 → এখানে 24 টি network address and 3 subnet address but device will see it like 27 bits network address.
- # fixed Subnet → একটা - same amount এর subnet হবে - and they will get same amount of ip, more waste of IP.
- # Variable Subnet → একক network এ ~~করা~~ amount of host basis এ একক - ~~করা~~ - subnet assign করে, less waste of IP.



Slide diagram



* সব সব sequentially - সব bit 01 - হল - তাই next bit এর value 10 (যেটা 1 করা - 200 নাহি)

01111111 127 total.

We can not take just one bit to create Subnet ip address, # we have to think about reducing ip waste

CSE 496

7. edi → 7. দ্বি-রেজিস্টার প্রকল্পে হয়,

- # There are two types of AT and Intel.
- # 32 bits set - 64 bits reset - 64 bits
- # Striped vs non striped in C file.

- # GUIDATA
- # GAD3
- # GADbolt
- # DOGBOLT

[CSE 421]

Every device that connects to a network needs an IP Address

Assigning IP address

1. Static → নিজেসব IP set করা, Router, server, printer will have static IP address.

2. Dynamic → Moving device এর dynamic address পাওয়া

Router DHCP দিবে না

Dynamic allocation → নির্দিষ্ট সময়ের জন্য IP দেয় device to

একটি device কোন ip নেই

1. That device will broadcast the message to the every device on a network.

2. That device এর reply দেয় - আমার থাকবে, আমি DHCP server এর reply দেয় - আমার থাকবে, particular device এর message পাঠাবে - এর unicast use করে,

3. Device IP address পাঠাবে - এর confirmation (প্রমাণ) - নির্দিষ্ট device এর 'I am using it'.

4. DHCP server - এর - আমার - এর - আমার - এর - 'you use it bro'

Net → why this message is in broadcast?

→ If there is multiple dhcp server, the device letting everyone know that it is using that dhcp server

→ minus acknowledgement → lease update
lease as time expires and
dhcp will send the device IP address, then dhcp
minus ack.

→ DHCP relay
→ Broadcast message for helper address
dhcp server and

NAT

Private IP address

→ ~~private~~ private address to public

Port mapping - mapping private and public
address

$$x^2 = (x)^2 \quad x \perp$$

Dynamic Routing $\rightarrow$ Software $\rightarrow$ best path $\rightarrow$ $\rightarrow$

Autonomous System

Router খুলে নিজে-
Network add. এর
- যা জানে যেটার
- এবং IP, subnet,
port এর মাধ্যমে

① Global $\Rightarrow$ All time shortest path फिर
 ② Local $\Rightarrow$ एक ही topology में idea नहीं

Distance vector $\rightarrow$ belmford algorithm, Neighbouring as perspective.
 Later goal quest, inefficient Algorithm

Same routing protocol use - একটি - হয় সব Router-এ

Same routing protocol use - 25 Router 20

Link state → [Process very important] because it is used in distributed systems.

- Example: Link state protocol - used in OSPF network.

- It is used in many networks.

Distance Vector and Link State Same

2. Hello packet ~~not~~ ^{is} sent at certain time or in two work $\rightarrow$ build connection and check if the connection is alive

১. Lsp packet A connected Network device নিচে Network device packet পাঠায়, Network device packet, Network device Hello packet এর থেকে বড়- হয়, Network device packet এর connected device Network device Network device hello packet এর reply দিয়ে।

4. এটা - সুবিধা - হলো Vector এবং - সারণী - হুদাই table
এবং - algorithm চালানোর না - ভাঙে একবারই Run এবং -
আব - সিদ্ধ ভাবনাই পারবে - যখন router table একবার
বিশেষ update হবে

5. $\frac{7}{10}$ CPU and Memory utilization long term & short term & vector distance vector

Link Vector কোন neighbour track-এর distance

হোটে কোল্ট topology এর জন্য Distance Vector Use ~~০৫৫~~ ১৬০
~~১৬০~~ ১ better Otherwise Link state better anyway.

9 —> Directly connected RT, statically connected, তারে-
কানেক্ট করা হয়েছে।

$G \rightarrow$ directly connected.